R USERS GROUP · INVITED TALK · OCTOBER 2026

# Agentic R

Using agentic systems to efficiently write, review and produce reproducible code



**David E. Rebellón-Sánchez**, MD, MSc, PhD(c)
London School of Hygiene & Tropical Medicine

HANDS-ON MATERIALS

github.com/derebellon/agentic-r-workshop

AGENDA

# What we will cover today

1. **What agentic AI is**
   From a model that answers to an agent that acts, uses tools and checks itself.

2. **A short history: from autocomplete to agents**
   How coding assistance evolved, and where R fits in.

3. **Agents inside R, and running them locally**
   ellmer, tool calling, and free local models with Ollama.

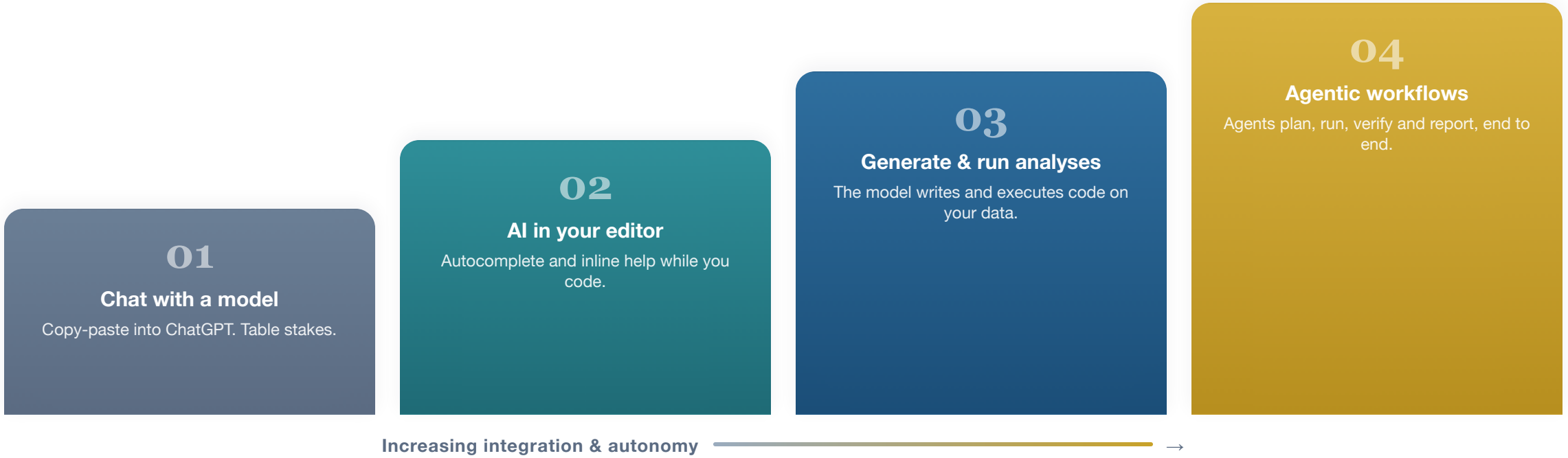
4. **Privacy with sensitive data**
   How to do all of this without your patient data ever leaving your machine.

5. **Live demo**
   An agent that builds a Table 1 and fits a model, on real-looking data.

FROM CHATTING TO DELEGATING

# Four ways to work with AI

Not a ranking of people, a spectrum of how deeply AI fits into your work. Each step adds integration and autonomy, and all of them are useful.



HOW WE GOT HERE

# From autocomplete to coding agents

Help with writing code has climbed the same ladder, from guessing the next word to running whole analyses for you.




### Autocomplete

The editor suggests the next token or function name, RStudio completion, IntelliSense. It guesses; it does not understand.

### LLMs generate code

Copilot, ChatGPT. You describe it in words and get a function or a whole script, but it lives outside R, in a chat window.

### LLMs inside R

Packages like **ellmer** (Posit) let you chat with any model **from R** and give it **tool calling**, so it runs your own R functions.

### Agentic systems

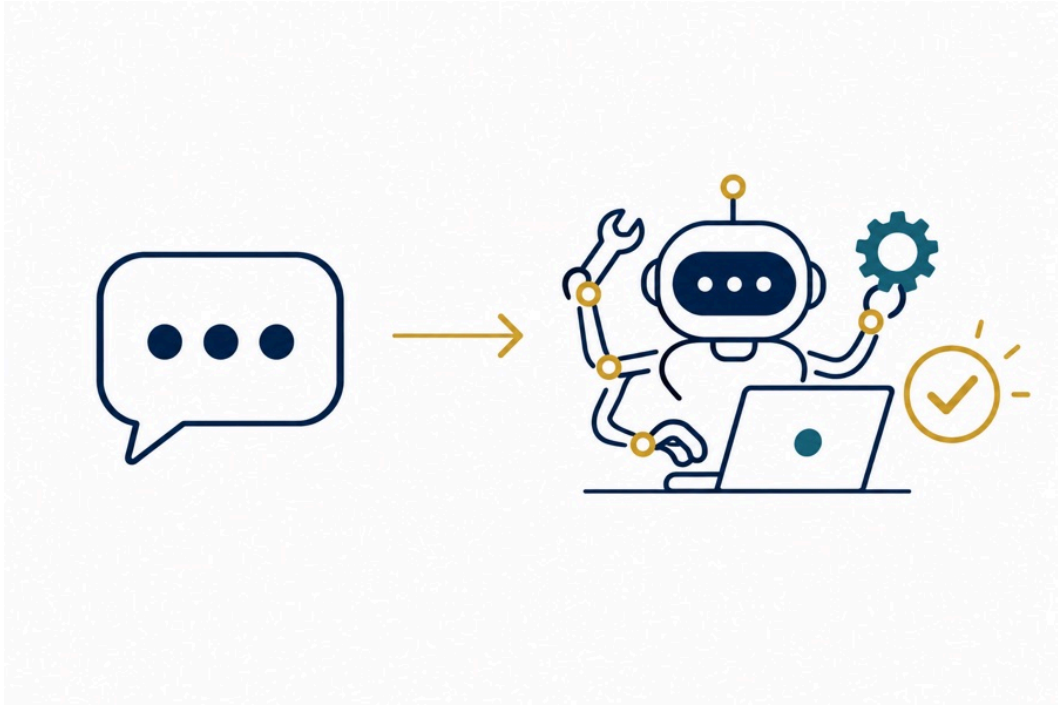
Claude Code and friends: agents that plan, run code, read the error, fix themselves and verify, a reproducible analysis end to end.

> The jump that matters for us: the model stops being a chat on the side and starts **acting inside your R session**, with your data and your functions.

PART 1

# What agentic AI is

The shift from a model that answers to an agent that acts, uses tools and verifies itself.



FOUNDATIONS

# From language model to agent

**FACT**

A **language model** (ChatGPT, Claude, Llama, Qwen) is excellent at predicting text: it answers, summarizes, translates and writes code.

**HOWEVER**

On its own it **does not act**: it does not run your R code, open your files or query your data, and it does not **verify** what it says. Sometimes it **hallucinates**.

**THEREFORE**

The **agent** appears: a model + tools + memory + an orchestrator (the "harness") that lets it act, use tools and check itself, with you supervising.

A chatbot **answers**. An **agent does tasks**: it loads your data, runs the model, reads the output, reviews it and reports the result back to you.

WHAT IT DOES

# What an agent does, day to day

**Review email**
Summarize, prioritize, draft replies.

**Write code**
R, Python, SAS: analysis and figures.

**Organize repos**
Structure projects in GitHub.

**Generate reports**
Reproducible, clear, professional.

**Transcribe**
Voice to structured text.

**Review literature**
Synthesis with verifiable citations.

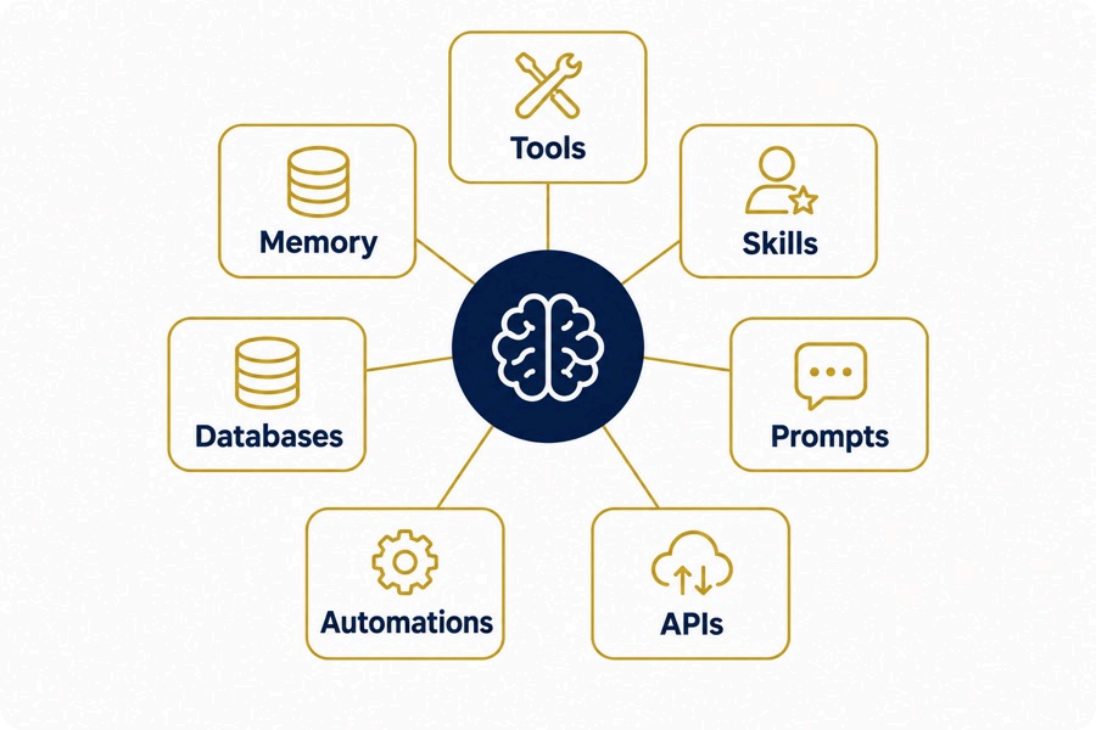
**Interpret omics**
Genomics, proteomics, microbiome.

**Draft science**
Protocols, methods, discussion.

**Autonomous agents**
They watch and report unprompted.

FOUNDATIONS

# Anatomy of an agent




- **Model**, the reasoning core. Open or closed, large or small, local or cloud.
- **Tools**, run your R code, read files, search literature, query databases.
- **Memory**, knows your project, your methods and your preferences.
- **Verification**, checks its own answers against the data and the output.
- **Human in the loop**, you approve, correct and decide. It accelerates; it does not replace your judgement.

START SMALL

# A minimal agent for R

You don't need databases, APIs, skills or long-term memory to be useful. For R, **three pieces** are enough, and the whole thing is about 50 lines.

### Model

A small local model (Qwen via Ollama), free and private, or Claude for more power.

### Tools

Your own R functions: load data, build a Table 1, fit a model. The agent calls them.

### Loop

The harness: perceive, reason, run a tool, verify, report. About 50 lines of R.

> In the workshop these are three files: **soul.md** (who it is) · **tools.R** (your R functions) · **agent.R** (the loop). Fill in the first two, and it is your agent.

FOUNDATIONS

# The harness and the working loop

The **harness** is the software that turns a model into an agent that works in a loop until the task is done: it **perceives**, **reasons**, **uses a tool** (runs code, reads your files), then **verifies and reports**, or repeats.

### Where does the harness live?

- **In an environment you own:** your own computer or server, with access only to the files you allow.
- **Sometimes you already use one:** every time you open the **Claude** or **ChatGPT** app, that is a harness running for you.

> In today's workshop the harness is about **50 lines of R**, so you can see every moving part.

```
# the loop, in words
repeat {
  perceive # read your request + state
  reason # decide the next step
  use_tool # run R, read a file
  verify # check the result
  if (done) report()
}
```

FOUNDATIONS

# Where can Agentic systems run?

You do not always need to send data to a public cloud. Think of it as a spectrum:

**More control & privacy** ——— **More convenience & power**

## Local

On your computer · Ollama

- Runs on your machine
- Data never leaves · free · offline
- Great for short, routine tasks

## Cloud

API / provider · Claude, GPT

- Runs in the provider's cloud
- Most powerful, long tasks
- Costs money · check retention

**Key message:** what matters is **where the model runs** and **what happens to your data**. The same R code works with either.

BE HONEST ABOUT THE LIMITS

# Risks, and how we manage them

### Hallucination

Confident, plausible, sometimes wrong. → Verify every claim against the source.

### Representation bias

Trained on uneven data; weaker on under-studied populations. → Validate against your gold standards.

### Over-reliance

Skills atrophy if you stop thinking. → AI accelerates; your judgement still decides.

### Reproducibility

Black-box steps are not science. → Log tools, parameters and versions; keep it auditable.

### Data exposure

Sensitive data in the wrong channel. → Anonymize, run local, or no-retention by contract.

### Cost & access

Not free, and unevenly distributed. → But local models are free; give access to the right people.

> None of these is a reason not to start. Each is **controllable by design**, and the biggest one, hallucination, is why a human stays in the loop.

HOW WE STAY HONEST

# Build disagreement in, keep a human in charge

A model can **hallucinate**, and tell you the story you *want* to hear. So we make the system **argue with itself**, and keep a human in the loop.



Two agents, on purpose: a **worker** that runs the analysis, and an **opposer** that hunts biases and limits, "*what would make this wrong?*"



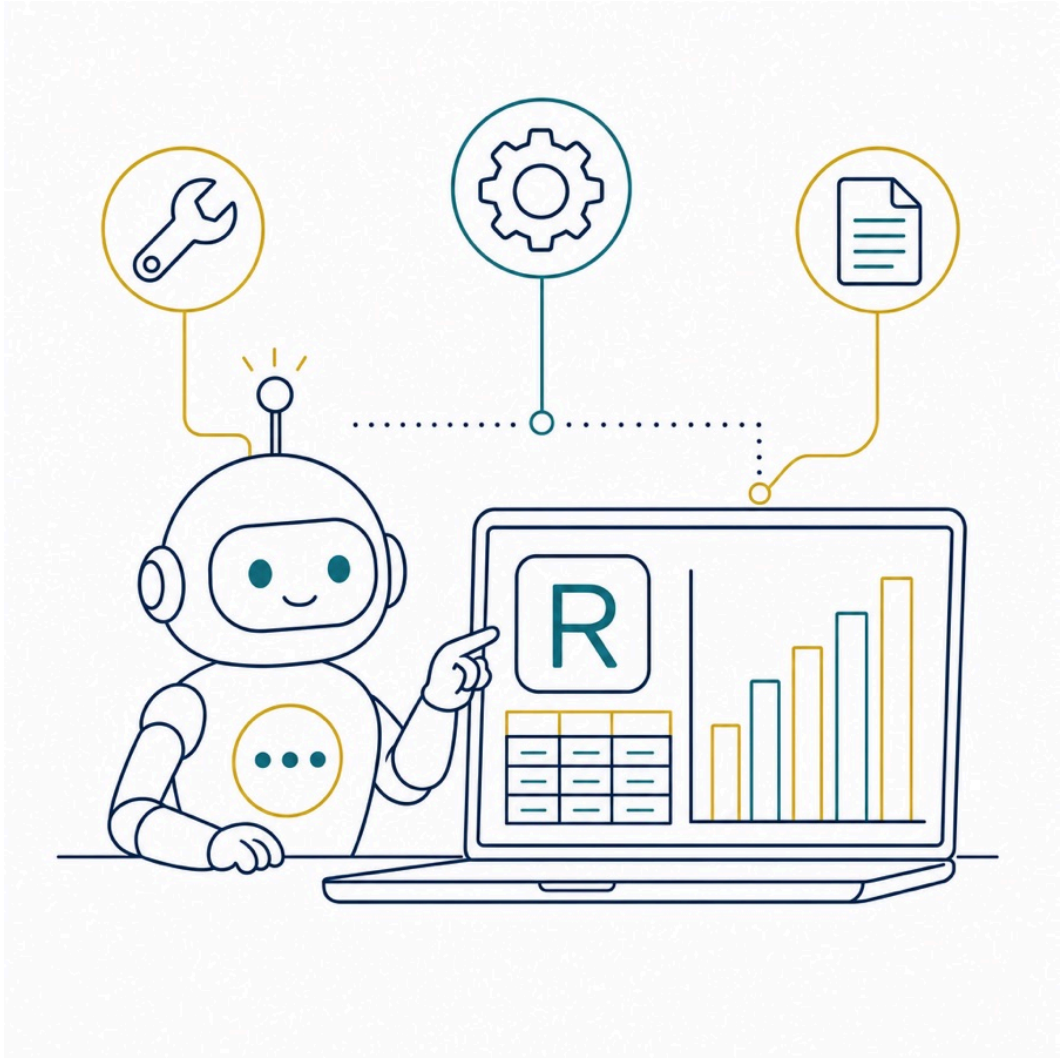
Then **you** check the findings against the literature and decide. The agent accelerates; **your judgement decides**.

> Ask the model for **two things**: the analysis you want, **and an adversary that tries to break it**. Agreement after a real fight beats a confident first answer.

PART 2

# Agents inside R

Chat with any model from R, give it your functions as tools, and run it on a free local model.



AGENTS INSIDE R

# LLMs inside R: the ellmer package

**ellmer** (from Posit) is the bridge: it lets you talk to **any** model from R, local or cloud, with one API.

- **Tool calling:** register your own R functions, the model decides when to run them. That is what turns it into an agent.
- **Structured output:** extract text into a typed table (great for messy clinical notes).
- **Any engine:** a local model with **Ollama** (free, private) or Claude in the cloud, swapped in one line.

> Around ellmer sit the editor assistants: **chattr**, **gander**, **btw**, the agent, right inside RStudio.

```r
# chat with a LOCAL model, free
chat <- chat_ollama("qwen2.5:3b")

# give it one of your R functions
chat$register_tool(tool(
  make_table,
  "Build a Table 1 with gtsummary"
))

chat$chat("Table 1 by treatment")
#> [tool] make_table(by = "treatment")
```

PART 3

# Privacy with sensitive data

How to do all of this without your patient data ever leaving your machine.



PRIVACY & COMPLIANCE

# How your patient data stays private

Your data stays on your computer. If anything is shared, it is only **de-identified** results.



**Your data**
Patient records, stay on your computer

→



**Local analysis**
R on your machine

→



**Results**
Tables, models, figures

→



**De-identify**
Remove names & identifiers



**Option 1 · Read locally**
A local model (Ollama) interprets the results. **Nothing ever leaves your computer.**



**Option 2 · Read in the cloud**
Only de-identified results travel. Use a provider that **does not store or train** on your data.

**Key message:** the sensitive data stays local; only de-identified results may be shared. With a local model, even that stays home.

HANDS-ON

# Try it yourself: a practical example

A tiny agent, **"Rho"**, that analyses a synthetic clinical cohort. Everything runs on your laptop.

- **Three folders:** an empty skeleton, the same one filled in, and the agent as a Claude Code project, same idea, three clothes.
- **You ask in plain English:** "Give me Table 1 by treatment, with p-values" and it runs `gtsummary` for real; then "which variables raise mortality?" and it fits a logistic regression.
- **Same code, two engines:** a free local model (Qwen via Ollama) or Claude, switched in one line.

> Clone it, run it, and make it yours: fill in its **soul** and its **tools** with the R libraries you like.

**github.com/derebellon/agentic-r-workshop**

# Thank you

Agents don't replace your judgement, they do the busywork around it.

**Model + tools + loop**
that is all an agent is

**ellmer + Ollama**
agents in R, free & private

**Local or cloud**
same code, your choice

> **Start small:** clone the workshop, run the local agent, give it one of your own R functions, and climb one rung of the ladder. **github.com/derebellon/agentic-r-workshop**